

Task 4: Setup and Use a Firewall on Windows

Objective

To configure and test basic firewall rules using Windows Defender Firewall and understand how firewalls filter network traffic.

Tool Used

- Windows Defender Firewall with Advanced Security
- Windows 11

System Configuration

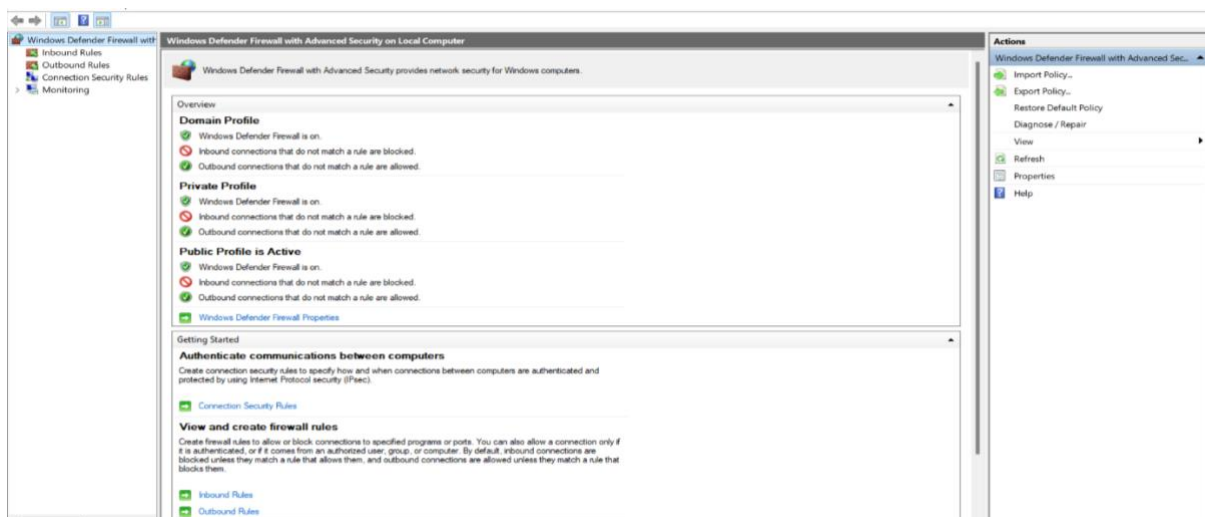
Parameter	Value
Operating System	Windows 11
Firewall Tool	Windows Defender Firewall
Rule Type	Inbound Rule
Protocol	TCP
Port Blocked	23 (Telnet)

Procedure

Step 1: Open Windows Firewall

1. Open Start Menu.
2. Search for "Windows Defender Firewall with Advanced Security".
3. Open the firewall management console.

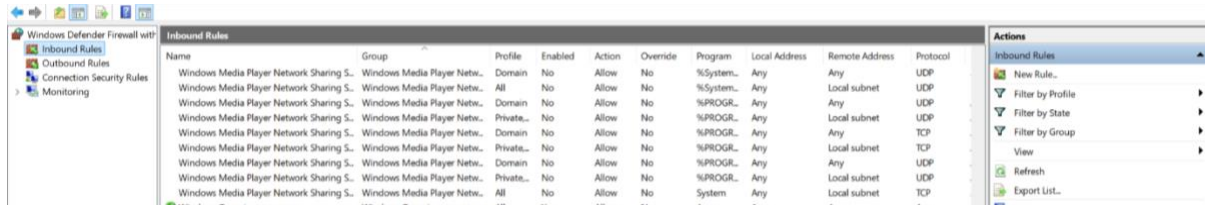
Screenshot 1: Firewall Dashboard



Step 2: View Existing Rules

1. Click on Inbound Rules.
2. Review the existing firewall rules.

Screenshot 2: Existing Inbound Rules



Step 3: Create New Firewall Rule

1. Click "New Rule".
2. Select "Port".
3. Click Next.

Screenshot 3: Rule Type Selection

Step 4: Configure Port

1. Select TCP.
2. Enter Port Number 23.
3. Click Next.

Screenshot 4: Port Configuration

Step 5: Block Connection

1. Select "Block the Connection".
2. Click Next.

Screenshot 5: Block Connection Option

Step 6: Apply Rule

1. Select Domain, Private and Public profiles.
2. Click Next.

Screenshot 6: Firewall Profiles

Step 7: Name the Rule

Rule Name:

Block Telnet Port 23

Description:

Temporary firewall rule for internship task

3. Click Finish.

Screenshot 7: Rule Creation

Step 8: Verify Rule

1. Locate the newly created rule.
2. Verify:
 - Enabled = Yes
 - Action = Block
 - Protocol = TCP
 - Port = 23

Screenshot 8: Rule Verification

Step 9: Test Rule

Attempted to verify the blocked Telnet port.

Command Used:

telnet localhost 23

Observation:

Telnet Client was not installed on the Windows 11 system. However, the firewall rule was successfully created and verified through the Windows Firewall interface.

Screenshot 9: Rule Details / Verification

Step 10: Remove Rule

1. Right-click the rule.

2. Select Delete.
3. Confirm deletion.

Screenshot 10: Rule Removal

Commands / GUI Steps Used

1. Opened Windows Defender Firewall with Advanced Security.
 2. Viewed existing inbound rules.
 3. Created a new inbound rule.
 4. Selected Port as the rule type.
 5. Configured TCP Port 23.
 6. Selected Block Connection.
 7. Applied the rule to Domain, Private and Public profiles.
 8. Verified rule creation.
 9. Removed the rule to restore the original firewall configuration.
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Observations

- Windows Firewall allows administrators to create rules that control network traffic.
 - Blocking Port 23 prevents Telnet connections from reaching the system.
 - Firewall rules can be applied to different network profiles such as Domain, Private and Public.
 - Rules can be enabled, disabled or removed as required.
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Result

Successfully configured and tested a Windows Firewall rule to block inbound traffic on TCP Port 23 (Telnet). The rule was verified and later removed to restore the original firewall configuration.

Conclusion

Windows Defender Firewall is an effective security mechanism that filters incoming and outgoing network traffic based on predefined rules. Proper firewall configuration helps protect systems from unauthorized access and network-based attacks.

